Lesson 7.11 — Modeling Periodic Phenomena

Big idea: Anything that repeats — tides, daylight, sound, oscillation — can be modeled with a sinusoid. Pull the four parameters (amplitude, period, phase shift, midline) right out of the data.

Quick review

Identify amplitude: (max – min)/2.
Identify midline: k = (max + min)/2.
Identify period: the time it takes for one complete cycle. $b = 2\pi/\text{period}$.
Choose sine vs. cosine: use cosine when the cycle starts at a max; use sine when it starts on the midline ascending.
Use a known data point (like a peak time) to pin down the phase shift.

Worked example

Worked example. A Ferris wheel's height swings from 5 ft to 65 ft, completing a cycle every 60 seconds, and starts at the bottom at t = 0. Write h(t).

Amplitude 30, midline 35. Period 60 → $b = 2\pi/60 = \pi/30$. Starts at min (bottom): use –cos. $h(t) = -30\cos((\pi/30)t) + 35$.

Warm-ups

1. A wave has max = 7 and min = –3. Find amplitude and midline.
2. A cycle takes 12 seconds. Find b for sin(bt).
3. If a max occurs at t = 5, what phase shift would put a cosine wave starting at its max at t = 5?

Practice

1. Tide at a beach swings from 1 ft to 9 ft, period 12 hours, high tide at noon. Write h(t) where t = hours after midnight.

2. Daylight in a city ranges from 9 hr to 15 hr over a year. Write D(t) (t in months) with longest day at t = 6 (June).

3. A pendulum's position is $x(t) = 5\sin(2\pi t)$. Find amplitude, period, and x(0.25).

Challenge

1. Average monthly temperature in a town swings from 30°F (Jan, t=1) to 80°F (Jul, t=7). Write a sinusoidal model T(t) where t is the month number (1–12). Use cosine.

2. A sound wave has amplitude 0.5, frequency 440 Hz (cycles per second), no phase shift, midline at 0. Write y(t) and find y(0.001).

Answer key — Lesson 7.11

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Amplitude $(7 - (-3))/2 = 5$. Midline $(7 + (-3))/2 = 2$.
2. $b = 2\pi/12 = \pi/6$.
3. $h = 5$ (the cosine peak naturally lives at $x = h$).

Practice

1. Amplitude $(9 - 1)/2 = 4$. Midline 5. Period 12 $\rightarrow$ $b = \pi/6$. High tide noon ($t = 12$) means cosine peak at 12: $h(t) = 4\cos((\pi/6)(t - 12)) + 5$.
2. Amplitude 3, midline 12. Period 12 $\rightarrow$ $b = \pi/6$. Max at $t = 6$: $D(t) = 3\cos((\pi/6)(t - 6)) + 12$.
3. Amplitude 5. Period $2\pi/(2\pi) = 1$. $x(0.25) = 5\sin(\pi/2) = 5$.

Challenge

1. Amplitude 25, midline 55, period 12 (months). $b = \pi/6$. Max at $t = 7$: $T(t) = 25\cos((\pi/6)(t - 7)) + 55$. (Check $t = 1$: $25\cos(-\pi) + 55 = -25 + 55 = 30$ ✓.)
2. $y(t) = 0.5\sin(2\pi \cdot 440\,t) = 0.5\sin(880\pi\,t)$. $y(0.001) = 0.5\sin(0.88\pi) \approx 0.5\sin(2.7646) \approx 0.5(0.3681) \approx 0.184$.